Computing Automorphism Group of Error-Correcting Codes

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Reference

Summary

The paper proposed a procedure to compute the automorphism group of an Error-Correcting Codes. The algorithm can handle codes over small fields ($q = 2, 3, 4$) with small $n = 20 \sim 40$. The proposed algorithm works the best if we can start with a invariant subset of codewords $W \subseteq \mathcal{C}$ under automorphism, with a rich enough structure and a mild cardinality. A usual starter choice would be minimum-weight codewords or next-to-minimum-weight codewords.
Goal: Compute the automorphism group

$$\mathrm{AUT}(V) = \{\, s \text{ monomial permutation} : V^s = V \,\}$$

for a linear code $V \subseteq \mathbb{F}_q^n$.

Invariant test function $F_W(m \text{ tuple}; \ell)$ (1)

The function F_W is an invariant used to rapidly test if an m-tuple of monomial coordinates $(\mu_1 j_1, \ldots, \mu_m j_m)$ can be the image of another m-tuple $(i_1, \ldots, i_m)$ under an automorphism of the code V.

Invariant test function $F_W(m \text{ tuple}; \ell)$ (2)

Projection (Π)

- For a vector w in the invariant set W, Π projects w onto the m specified coordinates, scaling by the inverse of the field elements μ_k:

$$\Pi(\mu_1 i_1, \cdots, \mu_m i_m; w) = (\mu_1^{-1} w_{i_1}, \cdots, \mu_m^{-1} w_{i_m}) \in GF(q)^m$$

The Invariance Map (XNUM)

- The map XNUM assigns the same integer to two non-zero vectors in $GF(q)^m$ iff one is a scalar multiple of the other ($v = \mu w$).
- The total number of non-zero scalar-multiplication classes

$$q^* = \frac{q^m - 1}{q - 1}.$$

Invariant test function $F_W(m \text{ tuple}; \ell)$ (3)

Invariant Function F_W

- F_W is a **histogram** (vector of counts).
- For a given m-tuple, $F_W(\mu_1 i_1, \ldots, \mu_m i_m; \ell)$ counts how many $w \in W$ satisfy
  1. w belongs to the ℓ-th quasi-invariant subset, W_ℓ.
  2. XNUM$(\Pi(\mu_1 i_1, \cdots, \mu_m i_m; w))$ corresponds to the ℓ-th scalar-multiplication class.

Invariance Check

If an automorphism s maps $(i_1, \ldots, i_m)$ to $(\mu_1 j_1, \ldots, \mu_m j_m)$, then the two count vectors must be identical:

$$F_W(\mu_1 j_1, \cdots, \mu_m j_m; *) = F_W(i_1, \cdots, i_m; *)$$

Refining the Invariant Set Starter (1)

From the starter invariant set, we want to further refine it to smaller quasi-invariant disjoint subsets.

- To provide the maximum practical structure by partitioning the initial quasi-invariant set W into a finer, non-trivial partition.
- Ideally, the resulting subsets W_k should approximate the **orbits** of W under $\mathrm{AUT}(V)$.
- A finer partition leads to a much more efficient search space pruning in the main CODEAUT algorithm.

Refining the Invariant Set Starter (2)

The Procedure of Partitioning (REFINE)

1. **Start:** Begin with the current partition $W = \bigcup W_i$.
2. **Select Test Case:** Pick a specific quasi-invariant subset W_k and a low-dimensional coordinate tuple $m \leq 3$.
3. **Check Invariance of Two Vectors:** For a fixed reference vector $w_a \in W_k$, compare $\forall w_b \in W_k$ using an invariant F_{W_k}.
4. **Split Rule:** If the chosen invariant can distinguish between vectors w_b and w_c within W_k then W_k is **not uniform** and must be split.
5. **Iterate:** The process is repeated until no further splitting can be achieved using any low-dimensional tuple.

Refining the Invariant Set Starter (3)

F_W with small tuple

- The algorithm tests the vectors $w \in W_k$ by looking at their **projections** onto these small tuples.
- Keeping the tuple length m small the algorithm can use an exact, non-randomized invariant that is computationally light enough to escape the $O(q^m)$ complexity of the full F_W for large m).

Constructing the Automorphism Group

- The construction uses the Schreier-Sims Algorithm based on the coordinate base $B = \{i_1, \ldots, i_t\}$.
- The invariant functions F_{W_k} is used as a sieve to prune the search tree for possible base images.
- The algorithm searches for all possible images $(\mu_1 j_1, \ldots, \mu_t j_t)$ of the base.
- If a partial image $(\mu_1 j_1, \ldots, \mu_m j_m)$ yields an invariant value $(F_{W_k}(\cdot))$ mismatch with the target $F_{W_k}(i_1, \ldots, i_m; *)$ the search branch is discarded.
- Successful images correspond to unique generators, which are added to the Strong Generating Set of AUT(V).